

 Team 5 — Software Engineering Project

Lakshmi Textile

A Smart Inventory Management System for Indian Textile Businesses
Built with Glassmorphic Design · Zero Installation Required

 HTML5

 CSS3

 JavaScript

 Chart.js

 LocalStorage

Press → or use arrow keys to navigate

The Challenge at Lakshmi Textile

Lakshmi Textile is an Indian textile shop struggling with zero digital tracking of inventory. All materials coming in and going out are managed through paper-based processes.

"We don't know how much stock we have, what's selling, or what we're losing."



Paper-Based Tracking

Manual registers — lost, damaged, or error-prone



No Stock Visibility

No real-time view of Godown vs. shop floor stock



Untracked Losses

Wastage, damage & theft go completely undetected



Manual Calculations

Pricing, margins & GST done by hand — prone to errors

Designing for Non-Technical Users

The workforce includes warehouse staff, storekeepers, and sales clerks with varying technical expertise.



Large Buttons

Touch-friendly, oversized controls with clear labels



Minimal Typing

Dropdowns, selectors & auto-complete instead of text entry



Visual Indicators

Traffic light: Green (OK), Yellow (Low), Red (Out)



Multilingual

English, Hindi & Telugu — instant switching



Error Prevention

Confirmation modals, undo buttons & stock validation



Role-Based Views

Different dashboards for Owner, Counter & Godown staff



Responsive Design

Works on PC, tablet & mobile screens



Zero Installation

Double-click HTML — works instantly on any computer

A Standalone Glassmorphic Web Application

We built a fully offline, serverless web application that runs by simply opening an HTML file. No internet, no backend, no installation.

- Runs in any browser — Chrome, Firefox, Edge
- All data saved locally using Browser LocalStorage
- Export & Import JSON backups with one click
- Pre-seeded with sample Indian textile data

 English हिन्दी తెలుగు

Tech Stack



HTML5

Structure & semantic markup



CSS3 (Glassmorphism)

Frosted glass, blur effects, animations



Vanilla JavaScript

Business logic, CRUD, navigation



Chart.js (CDN)

Interactive data visualizations



Browser LocalStorage

Persistent offline data storage

Role-Based PIN Login System

Three distinct roles with separate access levels — each protected by a simple 4-digit PIN for ease of use.



Malik (Owner)

1 2 3 4

- ✓ Dashboard & Reports
- ✓ Billing & Stock
- ✓ Vyapaari & Calculators
- ✓ Financial Analytics



Counter Staff

2 2 2 2

- ✓ Dukaan Billing
- ✓ Stock Viewing
- ✗ No profit visibility
- ✗ No supplier access



Godown Staff

3 3 3 3

- ✓ Stock Manager
- ✓ Order Receiving
- ✗ No billing access
- ✗ No financial data

Owner's Business Dashboard

Real-time financial overview with animated metric cards, interactive charts, and intelligent alerts.

₹ 45,193

Lifetime Earnings

₹ 45,193

This Month Sales

₹ 15,930

Net Profit



Sales vs Expense Trend

Dual-axis area chart comparing monthly revenue against purchase expenses over 6 months — spots spending patterns.



Stock Value Distribution

Donut chart breaking down capital tied in each category — Sarees, Thans, Kurtis, etc.



Monthly Target Rings

SVG progress circles showing sales target achievement, profit margin health & stock utilization.



Low Stock Alerts

Color-coded alert cards for items below reorder level — one-click option to place orders immediately.

Dukaan Billing

(Counter POS)

A touchscreen-friendly Point of Sale interface for counter staff. Select category, pick item, set quantity, and bill — all in seconds.

1. Select Category

Tap visual tiles: Saree, Kurta, Than, Lungi, Dupatta, Accessories

2. Choose Item

Auto-filtered dropdown showing only items with available stock

3. Set Quantity

Large + / – buttons for meters or pieces. Supports decimals (0.5 m).

4. Complete Sale

Bill auto-calculates subtotal + 5% GST. Stock deducted instantly.

Sample Bill Output

Kanjivaram Silk Saree × 2 ₹ 9,600

Cotton Than × 10 m ₹ 1,300

Subtotal ₹ 10,900

GST (5%) ₹ 545

Total ₹ 11,445

Godown & Dukaan

Stock Manager

Complete inventory control with real-time adjustments, transfers, and wastage logging.



Add New Items

Name, category, quantity (Godown + Dukaan), buy/sell price, rack location & reorder threshold.



Quick +/- Adjustments

Inline buttons on every row for instant stock corrections — no modal needed for small changes.



Godown → Dukaan Transfer

Select item, specify quantity & transfer stock from warehouse to shop floor with one click.



Kharaab Maal (Wastage)

Log damaged or scrapped items with reason. Auto-deducts from stock and records financial loss.



Search & Filter

Instant search by item name, fabric type, or color. Filter by location or low stock status.



Traffic Light Status

Green = Normal, Orange = Kam Stock (Reorder!), Red = Maal Khatam (Out of Stock).

Supplier Hub & Order Tracker

A complete supply chain management module. Contact suppliers via WhatsApp, place orders & track deliveries through a visual pipeline.



- **Auto-Inward:** When 'Maal Mila' is clicked, stock auto-adds to Godown
- **WhatsApp Integration:** Pre-filled Hindi order messages sent to supplier
- **Spend Analytics:** Pie chart showing money spent per Vyapaari

📁 Vyapaari Directory

Name, city, phone, email, specialty, GSTIN — stored locally with one-click WhatsApp contact.

📝 Khareedi Order Creation

Select Vyapaari → Item → Qty → Cost → Expected Date. Total cost calculated live.

📊 Order History Charts

Horizontal bar chart of last 10 orders color-coded by status + supplier spend pie chart.

Smart Textile Business Calculators

Four purpose-built calculators that solve daily math problems for textile shop owners.



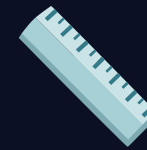
Kamaai Calculator

Profit Margin

Input: Buy Price + Overhead + Margin %

Output: Selling Price, Net Profit (₹), Markup %, GST-inclusive Price

Interactive slider for margin % that updates results in real time.



Than Yield

GSM to Meters

Input: Roll weight (Kg), fabric width (cm), GSM

Output: Estimated length in meters + yard conversion

Verify if suppliers sent correct fabric length without unrolling!



Mausam Predictor

Seasonal Demand

Input: Last month sales + Season dropdown (Diwali +80%, Monsoon -25%)

Output: Recommended order quantity with safety buffer

9 pre-set seasons from Indian festivals to school reopening.



Kharaab Maal

Wastage Loss

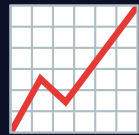
Input: Wasted meters + Cost per meter + Monthly sales

Output: Total loss (₹), Loss as % of sales, Severity gauge

Visual SVG gauge shows how badly wastage is eating into profits.

Interactive Data Visualizations

6 types of interactive charts powered by Chart.js, styled to match the glassmorphic design.



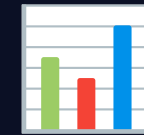
Sales vs Expense

Dual-line area chart with gradient fills — 6 month trend



Stock Donut

Category-wise stock value distribution with hover tooltips



Profit Bar Chart

Average margin % per category — identifies most profitable items



SVG Progress Rings

Monthly target %, profit health & stock capacity gauges



Supplier Spend Pie

Spend distribution across Vyapaaris — find biggest suppliers



Wastage Gauge

Semi-circular SVG gauge showing loss severity with color coding

Multilingual Interface

One-tap language switching between 3 languages. The selected language is remembered across sessions using localStorage.

🇬🇧 English

🇮🇳 Hindi

🇮🇳 Telugu

- 200+ translation keys covering every label, button & alert
- Uses data-i18n attributes for instant DOM updates
- Google Fonts with Noto Sans Devanagari & Telugu script support

English

हिन्दी

తెలుగు

Godown Stock

गोदाम स्टॉक

గోడౌన్ స్టాక్

Dukaan Stock

दुकान स्टॉक

దుకాణం స్టాక్

Vyapaari

व्यापारी

వ్యాపారి

Place Order

ऑर्डर माल

ఆర్డర్ చేయండి

Wastage

खराब माल

ఖరాబ్ మాల్

Net Profit

शुद्ध मुनाफ़ा

నికర లాభం

Calculator

कैलकुलेटर

లెక్కలు

Data Architecture & Backup System

All data is stored in the browser using localStorage with structured JSON. The app initialises with rich seed data on first launch.

Data Modules

 Stock Items	15 seeded
 Suppliers	5 seeded
 Orders	5 seeded
 Sales Records	10 seeded
 Wastage Log	2 seeded

One-Click Backup Export

Downloads a complete JSON file with all stock, sales, supplier & order data. Named with today's date for easy tracking.

Drag & Drop Restore

Upload a backup JSON file to restore all data — transfers between computers or recovers after browser history clearing.

Session Management

Login sessions use sessionStorage (auto-logout on browser close). Data persists in localStorage across sessions.

Project File Structure

```
team 5/  
├── index.html  
│   └── ← Double-click to run!  
├── css/  
│   ├── glass.css  
│   └── app.css  
├── js/  
│   ├── lang.js  
│   ├── data.js  
│   ├── auth.js  
│   ├── charts.js  
│   ├── calculators.js  
│   └── app.js  
└── presentation.html
```

8

Source Files

~140 KB

Total Size

200+

Translation Keys

0

Dependencies

🚫 Zero Backend

No servers, no databases, no build tools. Pure HTML + CSS + JS running entirely in the browser. Chart.js loaded from CDN for visualizations.

Thank You!

Lakshmi Textile's new inventory system empowers non-technical staff with a beautiful, intuitive & fully offline solution — built entirely with web technologies.



Ready to Demo?

Open the app live and show the audience!



Launch Lakshmi Textile App